EMPATHY MAP

SAYS

- It is difficult to recognize handwritten Tamil letters.
- Some characters look very similar.
- Manual identification takes more time.
- I want an easy and fast solution.

THINKS

- Will the system give the correct character?
- Can it recognize different handwriting styles?
- Is the model accurate enough?
- Will it be easy to use without technical knowledge?

DOES

- Writes Tamil characters on paper
- Studies Tamil language or documents
- Uploads handwritten images to the system
- Clicks the predict button to get results

In the past, Tamil texts were written by hand, making some characters difficult to read today. A system is needed to recognize these handwritten Tamil characters.

FEELS

- Confused when handwriting is unclear
- Frustrated with manual recognition
- Curious about AI-based solutions
- Happy and satisfied when correct prediction is shown

PAIN

- Difficulty recognizing handwritten Tamil characters
- Large number of characters with similar shapes
- Lack of automated Tamil OCR tools
- Time-consuming manual effort

GAIN

- Quick identification of Tamil characters
- Accurate recognition using CNN
- Reduced manual effort
- Easy-to-use and automated system

PROBLEM STATEMENT:

Handwritten Tamil character recognition is a challenging task due to the complex structure of the Tamil script and the large number of characters with similar shapes. Variations in individual handwriting styles, stroke thickness, and writing orientation make manual identification difficult and time-consuming. This creates problems for students, educators, and researchers who frequently work with handwritten Tamil documents.

At present, the recognition of handwritten Tamil characters is mostly done manually, which often leads to errors and requires significant effort. Existing digital solutions for Tamil handwritten character recognition are limited and do not provide reliable accuracy for real-world handwritten inputs. As a result, there is a need for an automated and efficient system that can accurately identify handwritten Tamil characters.

To address this problem, this project proposes a **Convolutional Neural Network (CNN)** based approach to recognize single handwritten Tamil characters from images. The system aims to reduce manual effort, improve recognition accuracy, and provide a simple and user-friendly solution for converting handwritten Tamil characters into digital form.